AYDIN ADNAN MENDERES UNIVERSITY
COMPUTER ENGINEERING DEPARTMENT

CSE 419 ARTIFICIAL INTELLIGENCE
ASSIGNMENT#1

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1.Introduction

- The problem in this project is to produce a solution by looking at the minimum waste and minimum cutting cost of the object of the given length with different optimizations and methods and finally the running times of the algorithms.

- The primary goal of the project is to minimize the amount of waste, after that, the cutting cost and working time are considered. Waste>Cut Cost>Time

- In the report, inputs were taken from two different users, one of the inputs is a difficult problem and the other is a very easy problem, we examined which method is better for these two cases.

2. Literature Review

- Hill Climbing (HC) is a simple and fast method. It only accepts better solutions at each step, so there is a high risk of getting stuck in local optimum.

- Simulated Annealing (SA) can avoid local minima by accepting bad solutions with a certain probability. It usually provides more stable results.

- Genetic Algorithm (GA) works on a population basis and scans the solution space with operations such as crossover, mutation, etc. It produces powerful results, especially in complex problems.

3. Algorithm Application Methods

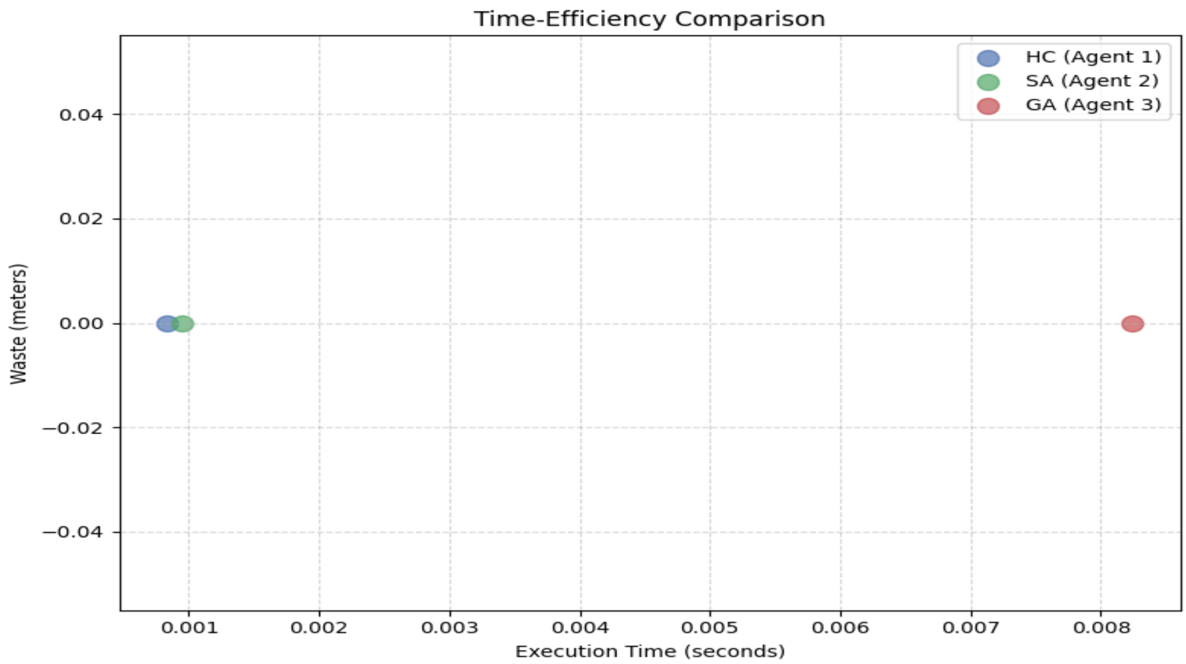
- 3.1. Single Algorithm Approach: In this method, each algorithm (GA, SA, HC) is run alone. Each one tries to solve the problem independently and their performance is evaluated separately. This approach is used to compare the individual success of the algorithms.

- 3.2. Collaborative Agent-Based Optimization: In this method, multiple agents work in parallel using the same algorithm. There may be information sharing or synchronization between agents. This approach is used to test whether multiple instances of the same method can jointly produce better solutions.

- 3.3. Hyper Meta-Heuristic Approach: In this method, different algorithms work together and can learn from each other or try to improve each other's solutions. The aim is to obtain higher quality solutions by taking advantage of the strengths of each algorithm.

4. Experimental Setup

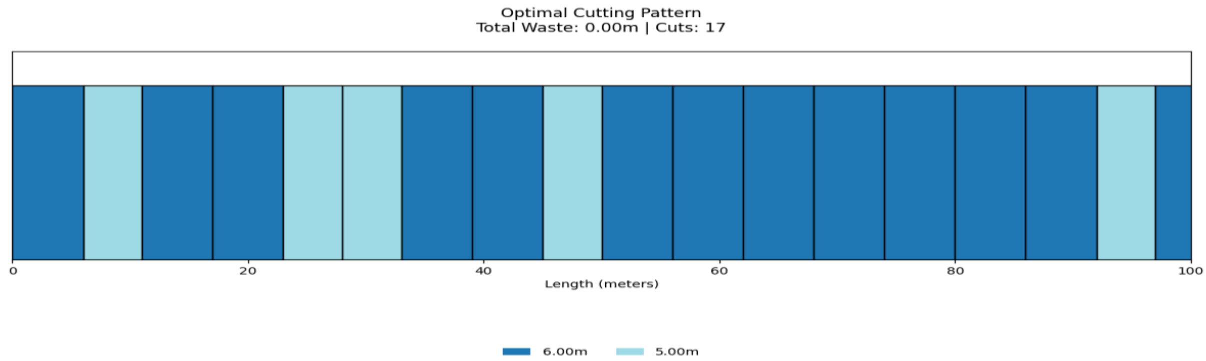
Time-Efficiency Comparison (Easy Problem)



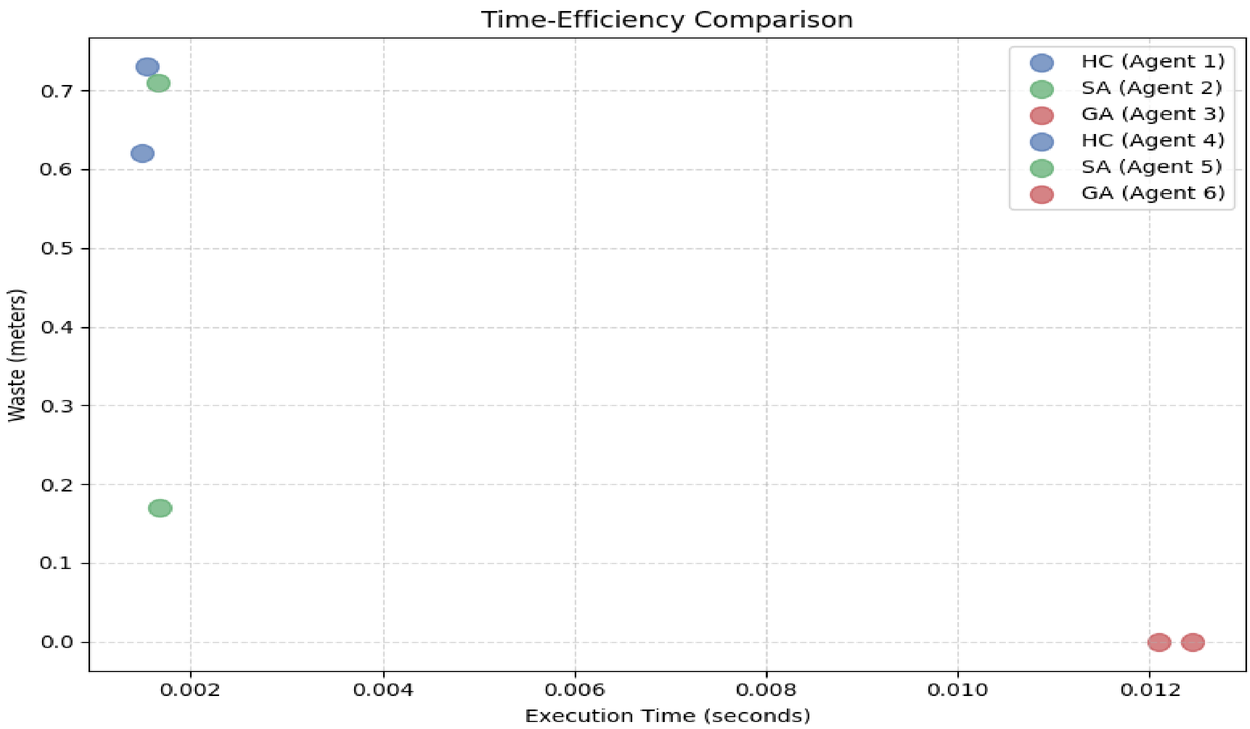
- Length:100 meters , Pieces: 4m- 5m-6m , 0m Waste

- By examining the image, we reached the following conclusion. The GA algorithm is not ideal for use in simple experiments because it requires too much running time for small experiments. Working speed: HC > SA > GA

- HC: Very fast because it only chooses the best neighbor. SA: Slightly slower because of randomness and temperature control. GA: Slowest because of operations such as population management, crossover, mutation.



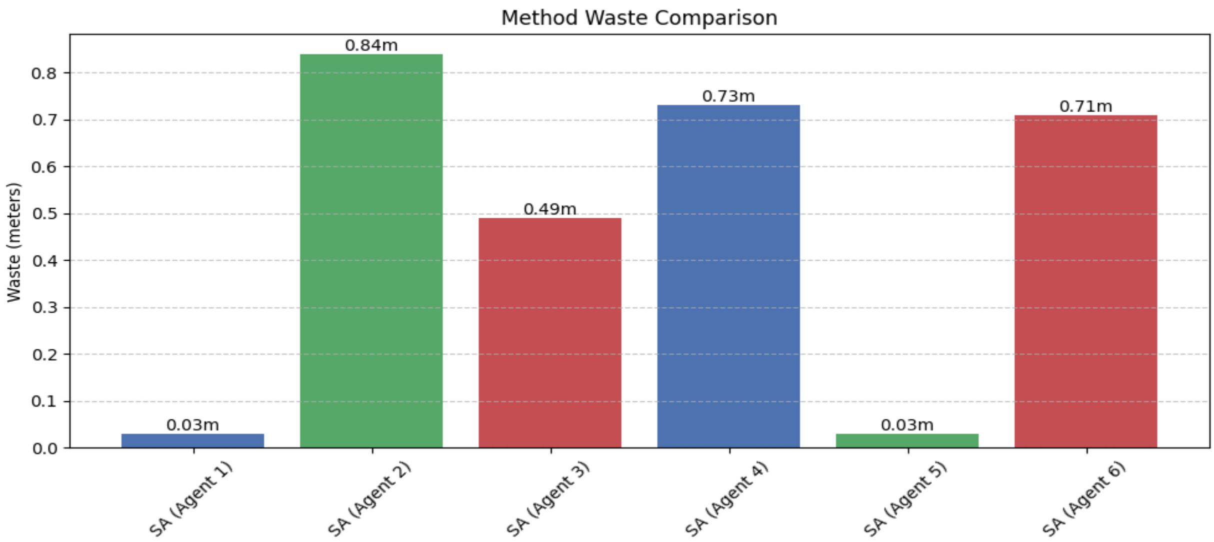
Time-Efficiency Comparison (Hard Problem)



- Length: 1234,5 meters , Pieces: 12.24m- 17.11m-13.37m , 0m Waste
- By examining the image, we reached the following conclusion. The GA algorithm is ideal for use in difficult problems because it gives less waste results for difficult problems, so the running time can be ignored.
- Solution Quality (Usually): GA > SA > HC

- HC: Very fast because it only chooses the best neighbor. SA: Slightly slower because of randomness and temperature control. GA: Slowest because of operations such as population management, crossover, mutation.

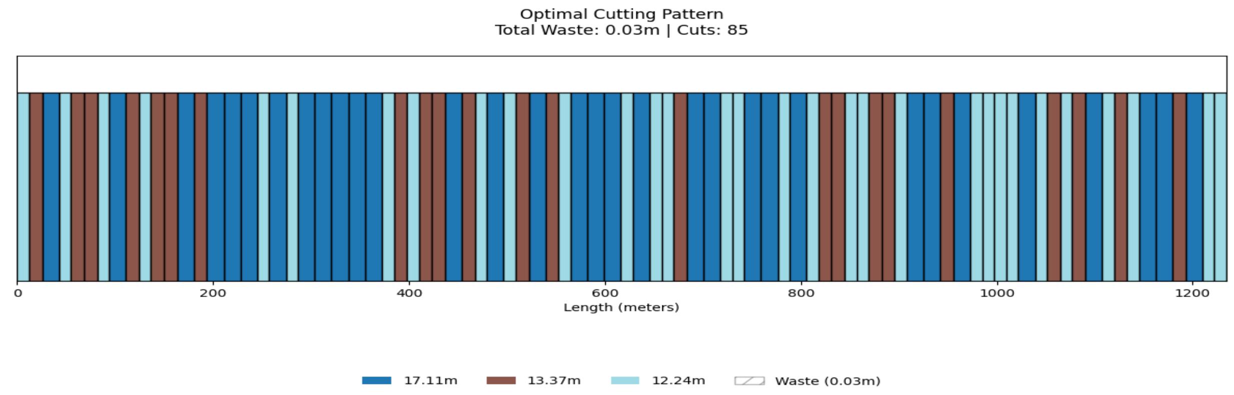
Collaborative Agent-Based Optimization



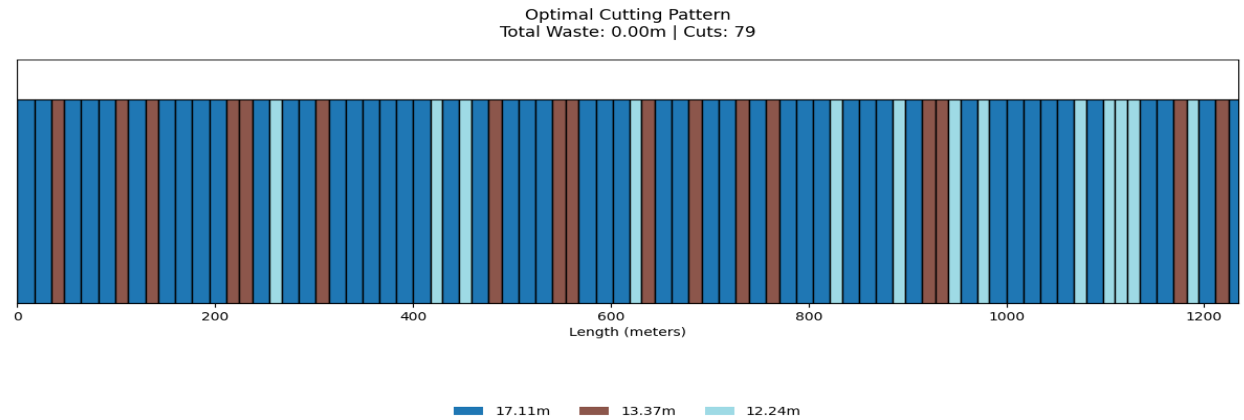
- Length: 1234,5 meters , Pieces: 12.24m- 17.11m-13.37m , 0.03m Waste

- The optimization method that uses more than one agent with the same algorithm determines the real performance by choosing the algorithm that is compatible with the difficulty of the problem (for example, using more than one agent using GA in a difficult problem allows you to reach a better solution).

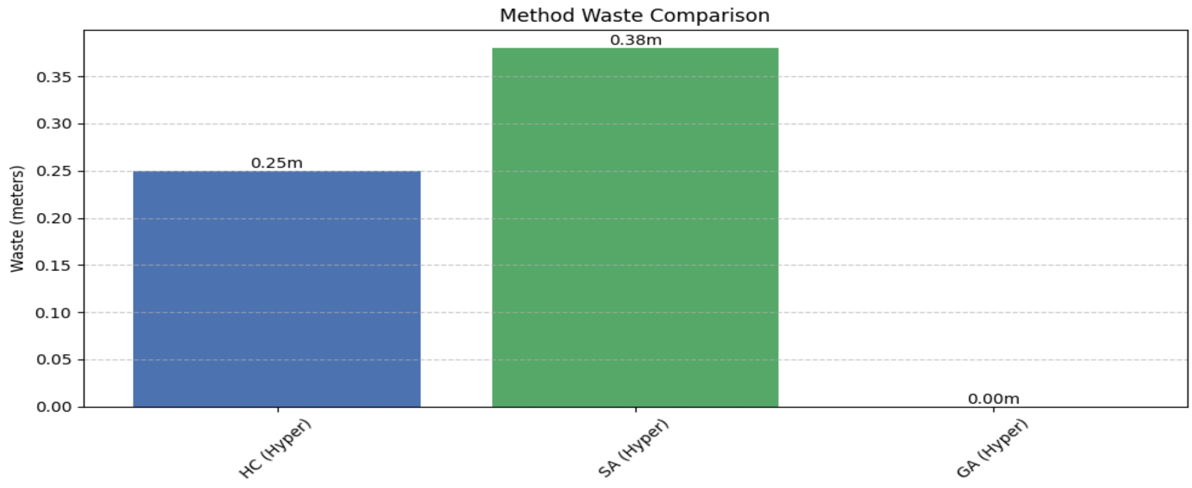
- At the same time, the running times of the GA algorithm are longer than the others, so SA or HC algorithms give better results than GA in easy problems.

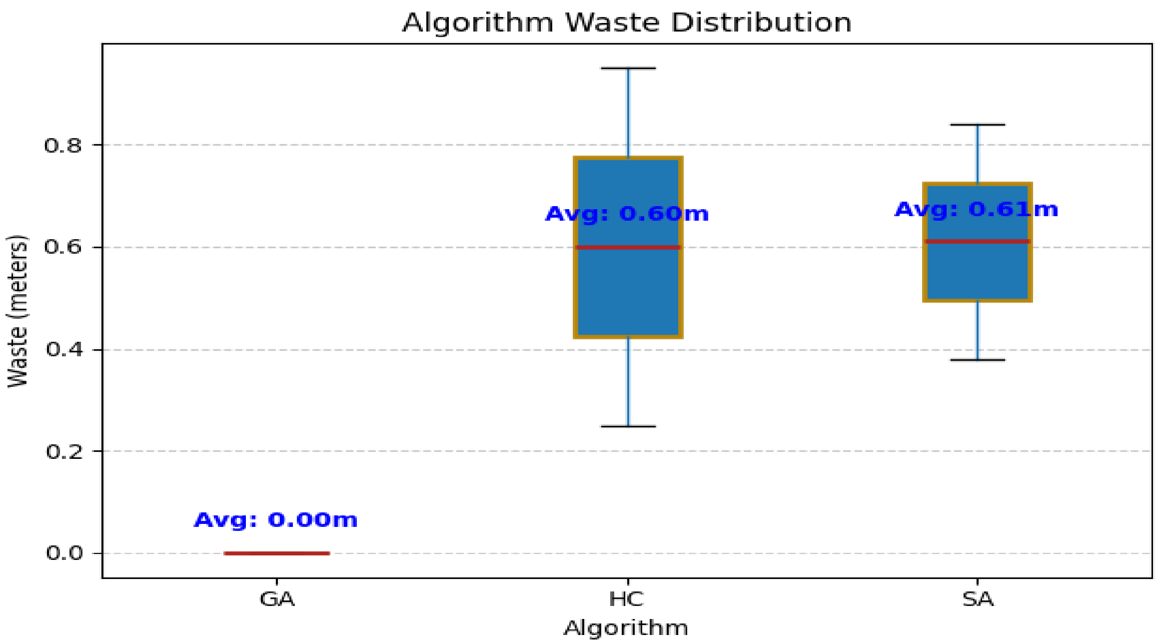


Hyper Meta-Heuristic Approach



- Length: 1234,5 meters , Pieces: 12.24m- 17.11m-13.37m , 0m Waste
- In cases where the difficulty of the problem is not specified, this approach may provide the best optimization to achieve the best results.
- However, using this method for easy problems causes unnecessary time expenditure and burden.





5.Results & Comparison

- Speed is important, approximate solution is sufficient | HC
- Local minima need to be avoided | SA
- Best solution needed, time is available | GA
- Solution Quality (Usually): GA > SA > HC
- Working Speed: HC > SA > GA

Criteria	Single Algorithm	Collaborative (Same Algorithm)	Hyper Meta-Heuristic (Different Algorithms)
Implementation Complexity	Low	Medium	High
Solution Quality (Generally)	Low–Medium	Medium–High	High
Execution Speed	Fast	Medium	Slow
Scalability	Limited	High	Very High
Best Use Case	Simple problems	Large problems, parallelizable	Complex, multi-faceted problems
Main Disadvantage	Risk of local optima	Coordination overhead	High computational cost and complexity

1. Single Algorithm Approach

When to use:

- When the problem structure is well understood.
- Suitable for small to medium-sized problems.
- When computational resources are limited.

Advantages:

- Simple to implement and maintain.
- Fast execution due to limited overhead.
- Easier to debug and analyze.

Disadvantages:

- Limited exploration of the solution space.
- May get stuck in local optima.
- Less robust for complex or dynamic problems.

2. Collaborative Agent-Based Optimization (Using the Same Algorithm)

When to use:

- When the problem is large and complex.
- Useful for parallel exploration using the same algorithm with different parameters.
- When agents can share information to improve collective performance.

Advantages:

- Allows distributed and parallel processing.
- Promotes diversity through collaboration.
- Increased chance of escaping local optima via shared learning.

Disadvantages:

- Communication and coordination between agents add overhead.
- May converge to similar solutions if diversity is not maintained.
- Requires careful synchronization to avoid redundant work.

3. Hyper Meta-Heuristic Approach (Using Different Algorithms Collaboratively)

When to use:

- For highly complex and multi-dimensional problems.
- When a single algorithm cannot efficiently handle the entire problem space.
- When combining the strengths of multiple algorithms is beneficial.

Advantages:

- Explores a wider solution space.
- Balances exploration and exploitation more effectively.
- Can adapt to various problem landscapes.

Disadvantages:

- Most complex to design and implement.
- High computational cost due to running multiple algorithms.
- Requires careful tuning and strategy management between different heuristics.